

Infosys Springboard

Project Report
on

ClimateScope

Visualizing Global
Weather Trends and Extreme Events

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PROBLEM STATEMENT

Understanding global weather patterns is challenging due to the complexity, volume, and fragmented nature of climate data. Existing solutions often lack interactive, user-friendly visualizations that allow effective analysis of seasonal trends, regional variations, and extreme weather events over time. There is a need for an accessible, data-driven platform that transforms daily-updated global weather data into meaningful insights. ClimateScope addresses this gap by providing interactive visual analytics to support climate awareness, informed decision-making, and further research into global weather dynamics.

ABSTRACT

Climate variability and the growing occurrence of extreme weather events present critical challenges for environmental assessment and infrastructure resilience worldwide. The ClimateScope project is a comprehensive, data-driven web application developed to analyze, visualize, and interpret large-scale climate data in order to support informed decision-making. The system integrates historical and near real-time weather datasets to examine key climatic parameters, including temperature, humidity, precipitation, wind speed, air quality, and extreme weather indicators across multiple geographical regions.

The project adopts a milestone-based development methodology encompassing data acquisition and preprocessing, exploratory data analysis, statistical evaluation, and interactive dashboard design. Using Python-based technologies such as Pandas, Plotly, and Streamlit, the application delivers intuitive and interactive visualizations, including time-series trend analyses, geospatial choropleth maps, heatmaps, and extreme event visual analytics. These visual components enable users to identify climate patterns, regional variations, and anomalies effectively.

The application is deployed as a scalable web-based platform on Streamlit Community Cloud, ensuring accessibility, interactivity, and real-time data exploration for researchers, policymakers, and stakeholders.

MILESTONE 1: DATA PREPARATION & INITIAL ANALYSIS

Tasks Performed:

- Collected the Global Weather Repository dataset from Kaggle.
- Set up the project development environment using Python and required libraries.
- Analyzed the dataset structure, data types, and key climate variables.
- Cleaned and preprocessed the dataset to handle anomalies and incomplete records
- Aggregated data to suitable temporal levels for further analysis.

Outcomes:

- Obtained a clean and well-structured dataset ready for analytical processing.
- Established a clear understanding of the dataset schema and variable definitions.
- Improved data reliability and consistency for accurate analysis.

	count	mean	std	min	25%	50%	75%	max	na_pct
temperature	3984.0	22.273553	8.680905	-24.900000	17.536613	24.420490	28.029810	45.000000	0.0
precipitation	3984.0	3.856591	6.261540	0.000000	0.070000	1.460000	4.712500	70.290000	0.0
wind_speed	3984.0	3.623921	1.779317	1.000008	2.306470	3.419830	4.622797	32.478964	0.0
humidity	3984.0	65.337959	21.579709	4.966667	51.169633	69.852688	82.042915	100.000000	0.0

```
Index(['station_id', 'country', 'month', 'temperature', 'precipitation',  
      'wind_speed', 'humidity', 'lat', 'lon'],  
      dtype='object')  
[19]:
```

	station_id	country	month	temperature	precipitation	wind_speed	humidity	lat	lon
0	'S Gravenjansdijk	Belgium	2024-05-01	16.000000	0.29	7.805618	82.000000	51.25	3.63
1	'S Gravenjansdijk	Belgium	2024-06-01	20.041176	2.52	4.666704	64.294118	51.25	3.63
2	'S Gravenjansdijk	Belgium	2024-07-01	21.250000	4.14	4.461345	63.071429	51.25	3.63
3	'S Gravenjansdijk	Belgium	2024-08-01	23.567742	2.37	4.674769	56.387097	51.25	3.63
4	'S Gravenjansdijk	Belgium	2024-09-01	18.736667	5.95	5.643564	70.533333	51.25	3.63

```
Data summary:  
<class 'pandas.core.frame.DataFrame'  
RangeIndex: 3984 entries, 0 to 3983  
Data columns (total 9 columns):  
#   Column              Non-Null Count  Dtype  
---  ---  
0   station_id          3984 non-null   object  
1   country              3984 non-null   object  
2   month                3984 non-null   datetime64[ns]  
3   temperature          3984 non-null   float64  
4   precipitation         3984 non-null   float64  
5   wind_speed           3984 non-null   float64  
6   humidity             3984 non-null   float64  
7   lat                  3984 non-null   float64  
8   lon                  3984 non-null   float64  
dtypes: datetime64[ns](1), float64(6), object(2)  
memory usage: 280.2+ KB  
None
```

MILESTONE 2: EXPLORATORY DATA ANALYSIS & VISUALIZATION

Tasks Performed:

- Conducted exploratory data analysis (EDA) to understand data distributions and trends.
- Examined regional and seasonal patterns across multiple countries.
- Identified extreme weather events such as heatwaves, heavy rainfall, and high wind conditions.
- Created preliminary visualizations including line charts, bar charts, heatmaps, and summary statistics.

Outcomes:

- Gained insights into global and regional climate behavior over time.
- Identified seasonal trends, anomalies, and extreme weather occurrences.
- Established key variables and metrics for detailed analysis.
- Formed the analytical basis for advanced visualizations and dashboard design.

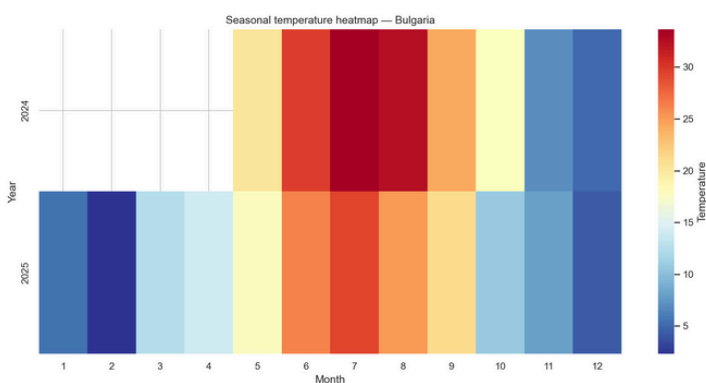


Fig 1:Bulgaria_seasonal_heatmap

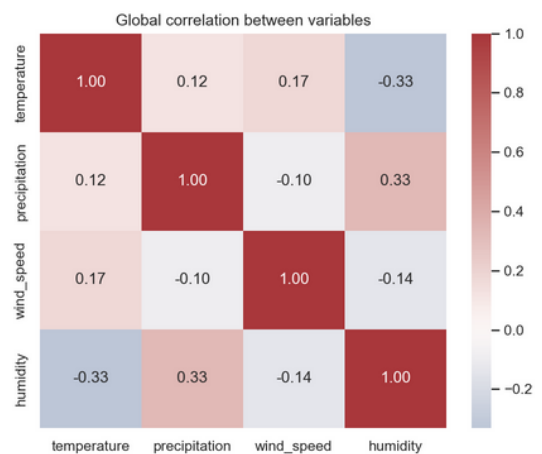


Fig 2:correlation_heatmap

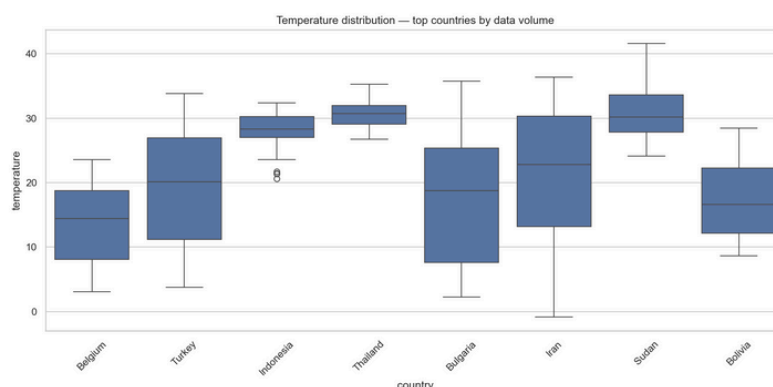


Fig 3:temp_boxplot_top_countries

MILESTONE 3: ADVANCED ANALYTICS

Tasks Performed:

- Developed static and interactive visualizations using Plotly within the Streamlit framework.
- Implemented user interaction components such as filters, sliders, and region selectors.
- Designed multiple dashboard pages to organize visual insights effectively.
- Refined visualization aesthetics through consistent color schemes, clear labeling, and improved layout.

Outcomes:

- Delivered a near-complete interactive dashboard prototype with core features implemented.
- Enabled dynamic data exploration across regions, time periods, and climate variables.
- Improved usability and visual clarity of the dashboard.

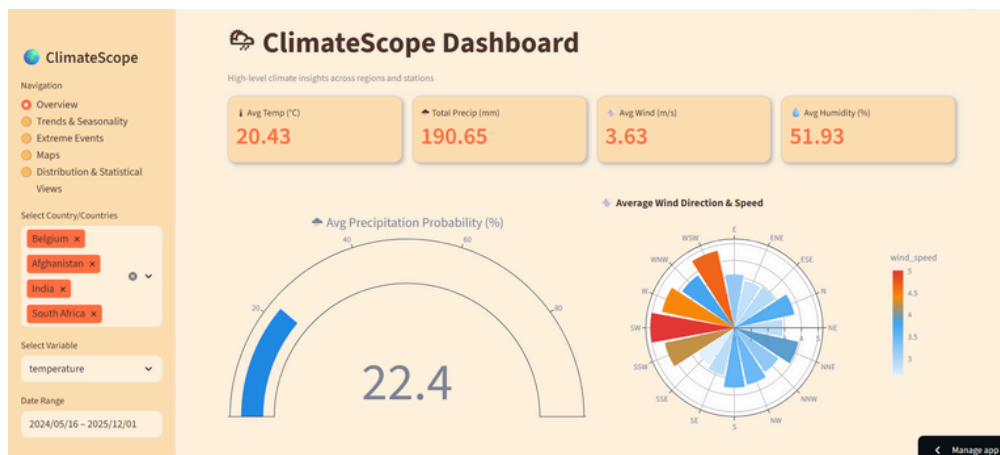


Fig 1:Dashboard

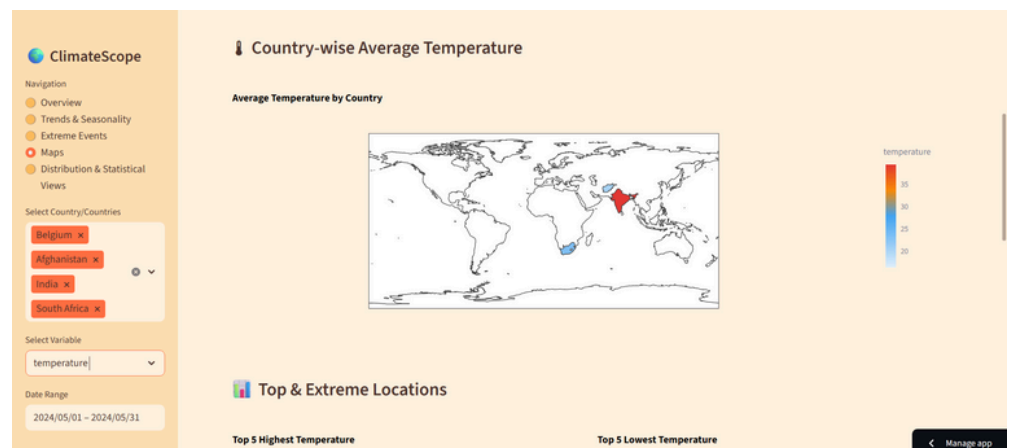


Fig 2:Map Page

MILESTONE 4: FINALIZATION, TESTING & REPORTING

Tasks Performed:

- Conducted comprehensive testing of the dashboard to validate functionality, data accuracy, and user experience.
- Verified the correctness of visualizations and interactive components across all dashboard pages.
- Documented the project in the final report.
- Deployed the dashboard as a web-based application for demonstration and accessibility.

Outcomes:

- Delivered a stable and fully functional climate analytics dashboard.
- Ensured reliability and accuracy of data-driven insights.

CONCLUSION

The ClimateScope project successfully demonstrates the application of data-driven analytics and interactive visualization techniques to analyze and interpret large-scale climate data. By integrating historical and near real-time weather datasets, the system provides meaningful insights into key climatic parameters such as temperature, humidity, precipitation, wind speed, air quality, and extreme weather events across multiple regions.

Through systematic data preprocessing, exploratory analysis, and advanced visualization development, the project delivers an intuitive and scalable web-based dashboard that enables effective exploration of climate patterns, regional variability, and anomalies. The inclusion of an Infrastructure Risk Management module further enhances the practical relevance of the application by translating climate indicators into actionable risk metrics for resilience planning.

Overall, the project highlights the effectiveness of interactive visual analytics in supporting environmental monitoring and informed decision-making, while providing a robust foundation for future extensions such as real-time data integration, advanced risk modeling, and enhanced regional analysis.

FUTURE ENHANCEMENT

- Integrate real-time weather APIs for live updates and forecasting.
- Incorporate machine learning models for anomaly detection and extreme event prediction.
- Add more granular geographic coverage, e.g., city-level data with high-resolution maps.
- Enhance user interactivity, including customizable dashboards and downloadable reports.
- Include climate comparison over multiple years to analyze long-term trends.
- Implement mobile-friendly and responsive design for broader accessibility.

REFERENCES

- Kaggle: Global Weather Repository Dataset.
Available at: <https://www.kaggle.com/>
- Plotly Technologies Inc. Plotly Python Open Source Graphing Library.
Available at: <https://plotly.com/python/>
- Streamlit Inc. Streamlit Documentation: Building Interactive Data Applications.
Available at: <https://docs.streamlit.io/>
- OpenWeather. OpenWeather API Documentation.
Available at: <https://openweathermap.org/api>
- Python Software Foundation. Python Language Reference and Documentation.
Available at: <https://www.python.org/>